

ACTIVE-BOX PRECONDITIONING FOR EQUISPACED-FOURIER KERNEL RIDGE REGRESSION

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Abstract. Kernel ridge regression with N samples usually requires a dense N -dimensional solve. For low-dimensional translation-invariant kernels, an equispaced Fourier expansion gives a fixed M -dimensional system with $M \ll N$ and $O(M \log M)$ matrix-vector products. Weak regularization can still make this system ill-conditioned. We identify Fourier modes whose diagonal kernel contribution is large relative to regularization and enclose them in a Cartesian box. Inside the box, the preconditioner uses either the inverse of the corresponding submatrix or a rank- q correction based on its leading eigenpairs; outside the box, it uses diagonal scaling. A condition-number bound separates the box correction, the diagonal approximation on the complement, and the coupling between them. We evaluate the complete binned EFGP pipeline against EFGP-CG on Synthetic and Winnebago with $N = 10^7$ – 3×10^8 . We report two proposed branches rather than one automatic route: the fastest active inverse and the fastest active-box EigenPro result, with full-grid EigenPro included as the latter’s limit. Synthetic favors the inverse branch, giving 5.36 – $8.52\times$ complete-pipeline speedup; Winnebago favors the eigenpair branch, giving 5.42 – $9.65\times$, with test accuracy shown beside every time. At $N = 3 \times 10^8$, total time falls from 15.328 to 2.862 seconds on Synthetic and from 27.748 to 2.901 seconds on Winnebago. We additionally report Nyström KRR, RPCholesky KRR, EFGP+Jacobi, and the full-grid extreme of our spectral family without hiding their raw speed-accuracy tradeoffs. Box/rank, data-set, and kernel studies show that the preferred branch changes with construction cost and data geometry. Finally, a controlled experiment gives every solver the same hashed $A\beta = \mathbf{b}$. Including active-set selection, preconditioner construction, and solve, the frozen default takes 1.481 seconds versus 6.525 seconds for CG, a $4.42\times$ paired-median speedup. The full-grid family member takes 1.276 seconds, a $5.13\times$ speedup, demonstrating the high-budget limit of the same spectral-preconditioning principle.

Key words. kernel ridge regression, preconditioning, Fourier methods, Toeplitz matrices, EigenPro

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1. Introduction. Kernel ridge regression (KRR) combines a flexible function class with a quadratic objective, but its usual data-space formulation contains a dense $N \times N$ kernel matrix. A direct solve costs $O(N^3)$ operations and $O(N^2)$ memory. Conjugate gradient (CG) removes the factorization, yet a dense kernel matrix-vector product still costs $O(N^2)$ [4, 17, 14, 31]. Large-scale KRR therefore needs both a cheaper representation of the kernel matrix and an iterative system that converges in relatively few steps.

For low-dimensional translation-invariant kernels, an equispaced Fourier expansion provides the first part. It replaces the data-space system by a weight-space system with M Fourier coefficients, where $M \ll N$ in the problems considered here. The coefficient matrix has the form $A = DGD + \lambda I_M$: D contains spectral weights and G is multilevel Toeplitz. The data enter only through one-time type-1 nonuniform fast Fourier transform (NUFFT) computations of the Toeplitz generator and right-hand side. After $O(N + M \log M)$ setup, a product with A costs $O(M \log M)$, with no direct dependence on N for fixed M , [15, 6]. This separation is especially useful when the number of observations is much larger than the Fourier grid.

The Fourier approximation and the iterative solution address different errors. The grid spacing and width determine the finite feature space and the approximation to the kernel, whereas the residual tolerance determines how accurately the resulting linear system is solved. Enlarging the grid may improve the kernel approximation, but it also increases M and the cost of each product. A preconditioner instead leaves

the exact solution of the fixed Fourier-discretized system unchanged and changes the work needed to reach the residual tolerance. Our algorithmic contribution addresses this second cost. The empirical comparison is broader and keeps the model-level and solver-level questions separate: end-to-end KRR pipelines are compared with their own setup costs, whereas the mechanism study fixes the Fourier model and varies only the solver.

The low-dimensional restriction is substantive. For a tensor grid with $2m + 1$ indices per coordinate, $M = (2m + 1)^d$, so neither the Toeplitz product nor the proposed box removes exponential growth with dimension d . The intended regime is small d , very large N , and a Fourier grid that fits in memory.

A fast matrix-vector product does not by itself give a fast solve. When the regularization parameter λ is small, the eigenvalues of A may span many orders of magnitude, and CG may take thousands of iterations. Each iteration remains inexpensive, but their accumulated cost can dominate both Fourier setup and prediction. The computational question studied here is therefore narrow: how can the Fourier system be preconditioned without discarding its fast Toeplitz product?

The diagonal structure of D gives a useful starting point. Fourier index j contributes $N|D_{jj}|^2$ to the diagonal of the unregularized system. The ratio $\rho_j = N|D_{jj}|^2/\lambda$ compares that contribution with the ridge term. A large ratio means that the unregularized diagonal contribution is not dominated by regularization, so the coordinate is worth treating accurately. This is only a selection rule: off-diagonal entries of G still couple the coordinates, so the ratio by itself does not prove that the remaining system is well conditioned.

We select the largest-ratio Fourier indices and enclose them in a Cartesian box B . The box has two practical roles. First, it gives a contiguous subgrid whose matrix A_{BB} can be obtained from the already computed Toeplitz data. Second, it leaves the complement available for inexpensive diagonal scaling. On B , we use either the inverse of A_{BB} or a rank- q correction built from its leading eigenpairs. The latter is the spectral preconditioner used by EigenPro; here it is used inside PCG, without EigenPro’s Richardson or stochastic training iteration. We denote the two choices by $P_{\text{inv}}(B)$ and $P_{\text{eig}}(B, q)$. The inverse gives the stronger correction when $|B|$ is moderate, whereas $P_{\text{eig}}(B, q)$ can treat a larger box at limited rank. Taking $B = J_m$ gives the corresponding full-grid cases, not separate methods.

The analysis makes explicit what this selection rule can and cannot ensure. The condition-number bound separates three effects: the accuracy of the operation on A_{BB} , the error of diagonal scaling on the complement, and the coupling between the two blocks. The ratio ρ_j helps control which coordinates remain outside the box, but the last two effects still depend on the Toeplitz matrix generated by the sample locations. This separation links the proof directly to the algorithm and explains why a larger box may reduce the iteration count while increasing total time.

1.1. Related work. Bochner’s theorem gives a spectral representation for a stationary positive-definite kernel [8, 36]. Random Fourier features draw frequencies from this density and obtain a finite feature model [24]. The construction used here instead evaluates the spectrum on an equispaced grid. This deterministic grid supports a controlled Fourier approximation and makes G Toeplitz [15, 6]. The present work leaves this Fourier approximation fixed and addresses the conditioning of its weight-space solve.

At the linear-solver level, diagonal (Jacobi) scaling is the simplest preconditioner for a positive-definite system [27]. For positive-definite Toeplitz systems, circulant

preconditioners have long been combined with conjugate gradients and FFT-based matrix-vector products [9]. Chan also proposed an optimal circulant approximation in the Frobenius norm [10]. These results do not apply directly to our system: although G is Toeplitz, the Fourier KRR matrix

$$A = DGD + \lambda I$$

is generally not Toeplitz because D is nonconstant. Our preconditioner instead preserves the fast Toeplitz product with G while using the Fourier weights in D to identify the coordinates that receive a stronger correction.

Another line of work reduces the model dimension through Nyström columns or inducing locations [37, 30]. KISS-GP places a regular interpolation grid in input space [38]; the grid here instead lies in frequency space. FALKON combines a sampled model with a preconditioned iterative solve [26, 22]. Randomized Nyström, sketching, and Cholesky constructions have also been used to precondition kernel systems [5, 13, 11, 12]. In particular, Nyström approximations can reduce the cost of spectral preconditioning while retaining much of its effect [3].

These methods typically construct data-space or inducing-point preconditioners from sampled kernel columns or sketches. In the end-to-end comparison, Nyström KRR and RPCholesky KRR are therefore treated as complete data-space KRR pipelines, including their own sampling, construction, and solve costs. Our active-box construction instead starts after the equispaced-Fourier model has been fixed. The resulting system has dimension M , and its Toeplitz generator is already available after Fourier setup. The active-box preconditioner therefore uses information already present in the Fourier system and requires no additional pass over the N training samples. Fourier-system adaptations of Nyström or RPCholesky are not used as the main end-to-end KRR baselines.

The location of the selected information changes the available computation. A data-space column corresponds to a sample location and requires kernel values against the data. Here a Cartesian frequency box gives a principal block that can be assembled from the Toeplitz generator after Fourier setup, without another pass over the N samples. Its subsequent costs depend on $|B|$, not directly on N . The price is that the score uses only D , N , and λ ; it cannot see the off-diagonal entries of the sample-dependent matrix G . The box is therefore a cheap proposal, not a substitute for data-dependent spectral information. The bound retains the missing coupling terms, and the experiments test a small set of boxes.

EigenPro introduced a spectral preconditioner that reduces the largest eigenvalues through a correction based on leading eigenvectors [20, 21, 1]. Recent work has reduced the cost of this spectral information using Nyström approximations [3], while EigenPro4 reduces the projection overhead of large kernel models through delayed projections [2]. These methods address spectral preconditioning and projection in general kernel training. Our $P_{\text{eig}}(B, q)$ restricts the spectral correction to a fixed Fourier submatrix. This changes its storage and application cost from $O(Mq)$ to $O(|B|q)$; when $B = J_m$, it gives the corresponding full-grid correction.

Standard preconditioning theory explains CG through the spectrum of the preconditioned matrix [16, 27]. A small condition number is not sufficient for lower wall time: the setup and repeated application of the preconditioner must also be cheaper than the iterations they save. Stage 1 therefore charges every KRR pipeline for its own setup and solve. Stage 2 holds A and $\mathbf{b}$ identical, reports the shared Fourier setup separately, and compares solver totals consisting of score selection (when applicable), preconditioner construction, and CG/PCG.

1.2. Contributions. The main contributions are as follows.

1. We distinguish a regularization-aware score threshold from a top- s selection budget. Exact threshold radii show polylogarithmic active-box growth for the squared exponential spectrum and algebraic growth for the Matérn spectrum before the finite grid saturates.
2. We combine $P_{\text{inv}}(B)$ or $P_{\text{eig}}(B, q)$ with diagonal scaling outside the box. A spectral bound separates the box correction, the diagonal approximation, and the interaction between them. A population Fourier-Gram formula further separates input-distribution correlations from finite-sample error in the last two terms.
3. We specify a two-stage evaluation that does not mix model setup with solver effects. Stage 1 first compares EFGP-CG with explicit inverse and active-box EigenPro branches—including the full-grid eigenpair limit—over 10^7 to 3×10^8 samples. Box/rank, data-set, and kernel changes use the same two-family reporting rule. Data-space Nyström, data-space RPCholesky, and EFGP+Jacobi then provide a secondary complete-KRR reference, with method-specific setup and solve accounting and explicit time-accuracy reporting. Stage 2 fixes A and $\mathbf{b}$, uses paired repeated timing, and compares CG, Jacobi and the proposed full-grid, active-inverse, and active-eigenpair family members by total time including score selection, preconditioner construction, and solve.

Section 2 derives the Fourier system and its fast product. Section 3 gives the two preconditioners, their spectral bound, and their costs. Section 4 reports the timing and accuracy results, and section 5 states the practical limits. Proofs and full algorithms are in the appendices.

1.3. Notation. The symbol $(\cdot)^*$ denotes conjugate transpose, and $\|\cdot\|_2$ denotes the Euclidean or spectral norm. For an index set B , A_{BB} is the principal submatrix indexed by B . For a Hermitian positive-definite matrix A , its condition number is $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$, and $\|v\|_A = (v^*Av)^{1/2}$.

2. Background.

2.1. Kernel ridge regression. Let $\{(x_i, y_i)\}_{i=1}^N$ be training data, with $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. For a positive-definite kernel k with reproducing kernel Hilbert space (RKHS) $\mathcal{H}$, KRR solves

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \sum_{i=1}^N (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}}^2.$$

We use the unnormalized KRR convention: the data-fit term is not divided by N . Thus a fixed λ , such as $\lambda = 0.1$, remains a fixed diagonal regularization level as N changes, while its strength relative to the $O(N)$ unregularized Fourier system decreases. Under the averaged-loss convention, the equivalent coefficient would be λ/N . The representer theorem gives $\hat{f}(x) = \sum_{i=1}^N \alpha_i k(x, x_i)$. With $\mathbf{K}_{ij} = k(x_i, x_j)$, $\mathbf{y} = (y_1, \dots, y_N)^\top$, and $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_N)^\top$, the coefficients satisfy $(\mathbf{K} + \lambda \mathbf{I}_N)\boldsymbol{\alpha} = \mathbf{y}$. This is an $N \times N$ dense system [4, 17, 28, 25].

2.2. Weight-space KRR. Suppose the kernel is approximated by M features:

$$k(x, x') \approx \tilde{k}(x, x') = \sum_{j=1}^M \phi_j(x) \overline{\phi_j(x')}, \quad \phi_j : \mathbb{R}^d \rightarrow \mathbb{C}.$$

Let $\Phi_{ij} = \phi_j(x_i)$, so $\tilde{\mathbf{K}} = \Phi\Phi^*$. For this fixed approximate kernel, the equivalent $M \times M$ system is

$$(\Phi^*\Phi + \lambda\mathbf{I}_M)\beta = \Phi^*\mathbf{y}, \quad \beta = (\beta_1, \dots, \beta_M)^\top \in \mathbb{C}^M.$$

The corresponding coefficients obey $\beta = \Phi^*\alpha$ and $\alpha = (\mathbf{y} - \Phi\beta)/\lambda$; these relations follow directly from the data-space and weight-space systems [25, 15, 6].

2.3. Equispaced Fourier expansion.

Kernel approximation. We consider translation-invariant kernels with $d \leq 3$ in the experiments. By Bochner's theorem, such a kernel has a nonnegative spectral density $\hat{k}$ such that

$$k(x, z) = \int_{\mathbb{R}^d} \hat{k}(\omega) e^{2\pi i \langle \omega, x-z \rangle} d\omega.$$

We approximate this integral on the grid $J_m = \{-m, -m+1, \dots, m\}^d$, which contains $M = (2m+1)^d$ frequencies. The resulting finite sum is [8, 15, 6]

$$\tilde{k}(x, z) = \sum_{j \in J_m} h^d \hat{k}(hj) e^{2\pi i h \langle j, x-z \rangle}.$$

The design matrix $\Phi \in \mathbb{C}^{N \times M}$ is

$$\Phi_{n,j} = \phi_j(x_n), \quad \phi_j(x) = \sqrt{h^d \hat{k}(hj)} e^{2\pi i h \langle j, x \rangle}, \quad j \in J_m.$$

For the kernels studied here, the approximation can reach high accuracy with $M \ll N$ [15, 6].

Fast matrix-vector products. Write $\Phi = FD$, where

$$\begin{aligned} F_{n,j} &= e^{2\pi i h \langle j, x_n \rangle}, & F &\in \mathbb{C}^{N \times M}, \\ D_{j,\ell} &= \sqrt{h^d \hat{k}(hj)} \delta_{j\ell}, & D &\in \mathbb{R}^{M \times M}, \quad j, \ell \in J_m. \end{aligned}$$

Because D is real and diagonal, the Fourier weight-space system is

$$(2.1) \quad A\beta = b, \quad A = DGD + \lambda\mathbf{I}_M, \quad G = F^*F, \quad b = DF^*\mathbf{y}.$$

Once h and m are fixed, this is the finite-dimensional system studied below. The preconditioner changes neither A , b , nor its exact solution. This does not identify the Fourier approximation with the original KRR problem. Because $G_{j,\ell} = g_{\ell-j}$, G is a d -level Toeplitz matrix determined by

$$g_r = \sum_{n=1}^N e^{2\pi i h \langle r, x_n \rangle}, \quad r \in J_{2m}.$$

The generator costs $O(N + M \log M)$ to compute with a type-1 NUFFT. Thereafter, padded FFTs apply G as a nonperiodic convolution in $O(M \log M)$ time [15, 7, 29]. The same type-1 NUFFT computes the right-hand side in (2.1), since $(F^*\mathbf{y})_j = \sum_{n=1}^N e^{-2\pi i h \langle j, x_n \rangle} y_n$. Thus setup costs $O(N + M \log M)$, and the product $Au = DG(Du) + \lambda u$ costs $O(M \log M)$, independently of N . With t_{iter} iterations, the complete solve costs $O(N + M \log M + t_{\text{iter}} M \log M)$ [15, 6].

2.4. Preconditioned conjugate gradient. Since A is Hermitian positive definite, CG solves $A\beta = b$ using only products with A . Its standard worst-case bound requires $O(\sqrt{\kappa(A)} \log(1/\varepsilon))$ iterations to reduce the relative A -norm error below ε [16, 27, 31].

Let P be a positive-definite approximate inverse. Preconditioned CG depends on both $\kappa(P^{1/2}AP^{1/2})$ and the cost of constructing and applying P . The iteration is given in Appendix B.

2.5. Selecting the active box. Since $|F_{n,j}| = 1$, (2.1) gives $G_{j,j} = N$ and $A_{j,j} - \lambda = N|D_{j,j}|^2$. For distinct $j, \ell \in J_m$, it also gives

$$A_{j,\ell} = D_{j,j}D_{\ell,\ell} \sum_{n=1}^N e^{2\pi i h \langle \ell - j, x_n \rangle}, \quad |A_{j,\ell}| \leq N|D_{j,j}||D_{\ell,\ell}|.$$

Here $D_{j,j} = \sqrt{h^d \widehat{k}(hj)}$. We rank the coordinates by $\rho_j := N|D_{j,j}|^2/\lambda = (A_{j,j} - \lambda)/\lambda$. This ratio is the diagonal contribution of Fourier index j , measured in units of the ridge term. A large value means that regularization does not dominate that coordinate, so omitting it from the accurate part of the preconditioner is risky. A small value also reduces its pairwise interactions with other small-score coordinates through the preceding bound. The ratio is therefore a plausible and inexpensive ranking rule, but it does not account for the sum of many off-diagonal interactions and is not by itself a conditioning guarantee.

For a threshold $\tau > 0$, define

$$(2.2) \quad S_\tau = \{j \in J_m : \rho_j > \tau\} = \{j \in J_m : N|D_{j,j}|^2 > \tau\lambda\}.$$

Thus $\tau = 1$ selects coordinates whose unregularized diagonal contribution exceeds the ridge term. For a selection budget s , let S_s contain the s largest-score indices. These rules have different roles because

$$(2.3) \quad \rho_j = \frac{Nh^d}{\lambda} \widehat{k}(hj).$$

On a fixed Fourier grid, the ordering used by S_s is independent of N and λ , whereas S_τ changes with N/λ . The top- s rule fixes the number of initially selected modes, although the enclosing box can contain more than s coordinates; the threshold rule is regularization-aware. We use S_τ as the primary definition and retain S_s as a budgeted alternative.

For either rule, write S for the selected set and enclose it in the smallest centered Cartesian box

$$(2.4) \quad B = \prod_{r=1}^d \{-b_r, \dots, b_r\}, \quad S \subseteq B \subseteq J_m.$$

We set $B = \emptyset$ when $S = \emptyset$. The box can include indices below the threshold, but it preserves the multilevel Toeplitz structure of G_{BB} . For a non-full box, define the induced threshold

$$(2.5) \quad \tau_B = \max_{j \in J_m \setminus B} \rho_j,$$

and set $\tau_B = 0$ when $B = J_m$. The strict inequality in (2.2) gives $B \supseteq S_{\tau_B}$. Consequently, the Fourier-system bounds in (3.2) also apply to a box obtained from a top- s selection. This containment does not assert that an arbitrary top- s box obeys the geometric size formula below.

PROPOSITION 2.1 (Growth of a threshold box). *Assume a unit-variance isotropic kernel and let $[u]_+ = \max(u, 0)$. The radii below are measured in the integer Fourier index j ; the corresponding frequency radii are hR_τ .*

For the squared exponential kernel

$$k_{\text{SE}}(x, z) = \exp\left(-\frac{\|x - z\|_2^2}{2\ell^2}\right),$$

define

$$(2.6) \quad R_{\text{SE}}(\tau) = \frac{1}{\sqrt{2\pi}\ell h} \sqrt{\left[\log\left(\frac{Nh^d(\sqrt{2\pi}\ell)^d}{\tau\lambda}\right) \right]_+}.$$

Then

$$(2.7) \quad S_\tau = J_m \cap \{j \in \mathbb{Z}^d : \|j\|_2 < R_{\text{SE}}(\tau)\}.$$

For the unit-variance Matérn parameterization

$$k_{\nu,\ell}(x, z) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\|x - z\|_2}{\ell}\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}\|x - z\|_2}{\ell}\right), \quad \nu > 0,$$

the Fourier convention in subsection 2.3 gives

$$(2.8) \quad \widehat{k}_{\nu,\ell}(\xi) = c_{\nu,d,\ell} (2\nu + 4\pi^2\ell^2\|\xi\|_2^2)^{-\nu-d/2}, \quad c_{\nu,d,\ell} = \frac{2^d\pi^{d/2}(2\nu)^\nu\Gamma(\nu + d/2)}{\Gamma(\nu)} \ell^d.$$

These transform formulas use the same 2π convention as [25, Sec. 4.2]; see also [6]. Define

$$(2.9) \quad R_{\text{Mat}}(\tau) = \frac{1}{2\pi\ell h} \sqrt{\left[\left(\frac{Nh^d c_{\nu,d,\ell}}{\tau\lambda}\right)^{2/(2\nu+d)} - 2\nu \right]_+}.$$

Then

$$(2.10) \quad S_\tau = J_m \cap \{j \in \mathbb{Z}^d : \|j\|_2 < R_{\text{Mat}}(\tau)\}.$$

If $S_\tau \neq \emptyset$, define

$$b_\tau = \max_{j \in S_\tau} \|j\|_\infty.$$

Its smallest centered box is $B_\tau = \{-b_\tau, \dots, b_\tau\}^d$, and

$$(2.11) \quad |B_\tau| = (2b_\tau + 1)^d, \quad b_\tau \leq \min\{m, \lceil R_\tau \rceil - 1\},$$

where R_τ denotes the applicable radius. If $R_\tau > m$, then $B_\tau = J_m$.

With d, h, ℓ, τ fixed and before truncation by J_m , the exact radii imply

$$(2.12) \quad R_{\text{SE}}(\tau) = O\left(\sqrt{\log(N/\lambda)}\right), \quad |B_{\text{SE}}| = O\left([\log(N/\lambda)]^{d/2}\right),$$

whereas

$$(2.13) \quad R_{\text{Mat}}(\tau) = O\left((N/\lambda)^{1/(2\nu+d)}\right), \quad |B_{\text{Mat}}| = O\left((N/\lambda)^{d/(2\nu+d)}\right).$$

Thus the SE box grows polylogarithmically in N/λ , while the Matérn box grows algebraically. On a finite grid either box can saturate at J_m ; the algebraic Matérn growth can lead to earlier saturation. If the kernel-approximation tolerance or lengthscale changes h , the exact radii in (2.6) and (2.9) should be used instead of these fixed-grid growth laws.

Let $B \supseteq S_\tau$ and $T = J_m \setminus B$. Then $\rho_j \leq \tau$ for every $j \in T$, so

$$A_{j,j} - \lambda \leq \tau\lambda, \quad |A_{j,\ell}| \leq \lambda\sqrt{\rho_j\rho_\ell} \leq \tau\lambda, \quad j, \ell \in T, \quad j \neq \ell.$$

These entrywise bounds do not control their combined operator norm. Section 3 therefore analyzes both the approximation on T and the coupling between B and T .

3. Active-Box Preconditioners. We partition the Fourier system in (2.1) into the box B selected in subsection 2.5 and its complement $T = J_m \setminus B$. The method treats these two parts differently: it uses a more accurate operation on A_{BB} , retains only diagonal scaling on T , and leaves products with the full matrix A unchanged.

3.1. Block-diagonal form. Let P_B approximate A_{BB}^{-1} . We use

$$P = \begin{bmatrix} P_B & 0 \\ 0 & \Delta_T^{-1} \end{bmatrix}, \quad \Delta_T = \text{diag}(A_{TT}), \quad (\Delta_T)_{j,j} = N|D_{j,j}|^2 + \lambda, \quad j \in T.$$

For a residual $r = (r_B, r_T)$, this means $Pr = (P_B r_B, \Delta_T^{-1} r_T)$. Constructing and applying Δ_T^{-1} each cost $O(|T|)$. The remaining cost depends on P_B .

3.2. Spectral bound. The preconditioner depends on its accuracy on A_{BB} , the error of diagonal scaling on T , and the coupling between B and T . Proposition 3.1 bounds their combined effect.

PROPOSITION 3.1 (Active-box spectral bound). *Let $A \in \mathbb{C}^{M \times M}$ be Hermitian positive definite, let $B \subseteq J_m$, and let $T = J_m \setminus B$. Let $\Delta_T = \text{diag}(A_{TT})$, $P_B \succ 0$, and $P = \text{diag}(P_B, \Delta_T^{-1})$. Define*

$$\begin{aligned} \mu_B^- &= \lambda_{\min}(P_B^{1/2} A_{BB} P_B^{1/2}), & \mu_B^+ &= \lambda_{\max}(P_B^{1/2} A_{BB} P_B^{1/2}), \\ \delta_T &= \|\Delta_T^{-1/2} A_{TT} \Delta_T^{-1/2} - I_T\|_2, & \gamma_{BT} &= \|P_B^{1/2} A_{BT} \Delta_T^{-1/2}\|_2. \end{aligned}$$

Every eigenvalue ξ of $P^{1/2} A P^{1/2}$ satisfies the first line below. If $\min(\mu_B^-, 1 - \delta_T) > \gamma_{BT}$, its condition number also satisfies the second line:

$$(3.1) \quad \begin{aligned} \min(\mu_B^-, 1 - \delta_T) - \gamma_{BT} &\leq \xi \leq \max(\mu_B^+, 1 + \delta_T) + \gamma_{BT}, \\ \kappa(P^{1/2} A P^{1/2}) &\leq \frac{\max(\mu_B^+, 1 + \delta_T) + \gamma_{BT}}{\min(\mu_B^-, 1 - \delta_T) - \gamma_{BT}}. \end{aligned}$$

COROLLARY 3.2 (Bounds for the Fourier system). *Let $B \supseteq S_\tau$ and $T = J_m \setminus B$. Then*

$$(3.2) \quad \begin{aligned} \delta_T &\leq \frac{\tau}{1 + \tau} \left\| \frac{G_{TT}}{N} - I_T \right\|_2, \\ \gamma_{BT} &\leq \sqrt{\frac{\tau}{1 + \tau}} \left\| \sqrt{N} P_B^{1/2} D_B \right\|_2 \left\| \frac{G_{BT}}{N} \right\|_2. \end{aligned}$$

The normalized blocks of G depend on the sampling design rather than on the kernel spectrum directly. The next result separates their population and finite-sample contributions, conditional on a fixed Fourier grid.

PROPOSITION 3.3 (Population Fourier Gram matrix). *Assume that $x_1, \dots, x_N$ are independent and identically distributed. For $r \in J_{2m}$, define*

$$(3.3) \quad \psi(r) = \mathbb{E} \left[e^{2\pi i h \langle r, x_1 \rangle} \right], \quad H = \mathbb{E} \left[\frac{G}{N} \right].$$

Then, for $j, \ell \in J_m$,

$$(3.4) \quad H_{j,\ell} = \psi(\ell - j), \quad \mathbb{E} \left| \frac{g_r}{N} - \psi(r) \right|^2 = \frac{1 - |\psi(r)|^2}{N} \leq \frac{1}{N}.$$

If $x_n \sim \mathcal{N}(\mu, \Sigma)$ with $\Sigma \succeq \sigma_x^2 I_d$ for some $\sigma_x > 0$, then

$$(3.5) \quad \psi(r) = \exp(2\pi i h \langle r, \mu \rangle - 2\pi^2 h^2 r^\top \Sigma r).$$

Let

$$(3.6) \quad a = 2\pi^2 h^2 \sigma_x^2, \quad \eta_d(a) = \sum_{r \in \mathbb{Z}^d \setminus \{0\}} e^{-a \|r\|_2^2}.$$

For every partition $J_m = B \dot{\cup} T$,

$$(3.7) \quad \|H_{TT} - I_T\|_2 \leq \eta_d(a), \quad \|H_{BT}\|_2 \leq \eta_d(a),$$

where the explicit lattice-sum bound

$$(3.8) \quad \eta_d(a) \leq \left[1 + \left(2 + \frac{1}{a} \right) e^{-a} \right]^d - 1$$

holds. Finally, for a fixed finite Fourier grid,

$$(3.9) \quad \left\| \frac{G}{N} - H \right\|_2 \rightarrow 0 \quad \text{almost surely as } N \rightarrow \infty.$$

Remark 3.4 (Uniform inputs). For the synthetic design $x_n \sim \text{Unif}([0, 1]^d)$,

$$(3.10) \quad \psi(r) = \prod_{s=1}^d e^{\pi i h r_s} \text{sinc}(h r_s), \quad \text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \quad \text{sinc}(0) = 1.$$

In particular, $h = 1$ gives $H = I_M$. For the kernel-dependent value of h used in an EFGP grid, (3.10) instead gives the exact population matrix. Thus the uniform synthetic experiment does not require a Gaussian-input assumption.

COROLLARY 3.5 (Population and sampling contributions). *Under the assumptions of Proposition 3.3, define*

$$(3.11) \quad \varepsilon_{TT} = \left\| \frac{G_{TT}}{N} - H_{TT} \right\|_2, \quad \varepsilon_{BT} = \left\| \frac{G_{BT}}{N} - H_{BT} \right\|_2.$$

If $B \supseteq S_\tau$, then

$$(3.12) \quad \delta_T \leq \frac{\tau}{1 + \tau} (\|H_{TT} - I_T\|_2 + \varepsilon_{TT}),$$

$$(3.13) \quad \gamma_{BT} \leq \sqrt{\frac{\tau}{1 + \tau}} \left\| \sqrt{N} P_B^{1/2} D_B \right\|_2 (\|H_{BT}\|_2 + \varepsilon_{BT}).$$

For Gaussian inputs, each population norm on the right is at most $\eta_d(a)$. For a fixed grid and a fixed partition, both sampling errors converge almost surely to zero. This last statement concerns sampling error; it does not assert a fixed-box limit when the threshold box itself changes with N .

Remark 3.6 (What the spectral bound establishes). Proposition 3.1 is a decomposition rather than a condition-number guarantee from the score alone. The interval $[\mu_B^-, \mu_B^+]$ measures the remaining error in the active-box operation, δ_T measures the error of diagonal scaling on the complement, and γ_{BT} measures the active-tail coupling omitted by the block-diagonal preconditioner.

The threshold enters the bound for δ_T through $\tau/(1+\tau)$ and the bound for γ_{BT} through its square root, but it does not by itself control the normalized blocks of G . Proposition 3.3 shows that these blocks contain a population correlation determined by the input distribution and a finite-sample error. The latter vanishes for a fixed grid as N grows, whereas the population term need not.

For nested boxes, increasing B cannot increase the induced threshold τ_B . It does not necessarily make the unweighted matrix $G_{TT}/N - I_T$ small, because neighboring Fourier indices can remain in T . Consequently, the score supplies a cheap candidate box, while μ_B^- , δ_T , and γ_{BT} describe whether that candidate gives an effective preconditioner. Measured construction and solve costs then determine which effective candidate is practical.

The next two subsections specify the two choices of P_B .

3.3. Inverse on the active box. For a sufficiently small active box, set $P_B = A_{BB}^{-1}$. The full preconditioner is

$$P_{\text{inv}}(B) = \begin{bmatrix} A_{BB}^{-1} & 0 \\ 0 & \Delta_T^{-1} \end{bmatrix}.$$

Here $P_B^{1/2} A_{BB} P_B^{1/2} = I_B$, so Proposition 3.1 has $\mu_B^- = \mu_B^+ = 1$. If $B = J_m$, this is the inverse of the full Fourier system; diagonal scaling on T then disappears. This limiting case is useful when the Fourier grid is small, but it does not change the definition of the method.

The block $A_{BB} = D_B G_{BB} D_B + \lambda I_B$ is obtained from the precomputed Toeplitz generator. Assembly and inversion cost $O(|B|^2 + |B|^3)$, and each application costs $O(|B|^2)$. Thus this choice is practical only when the box is moderate. Its advantage is that no rank must be chosen and the box contribution to the bound is exact. The reported implementation forms A_{BB}^{-1} from a Cholesky factorization; this construction is included in the preconditioner build time.

3.4. Leading-eigenpair correction on the active box. For a larger active box, let the eigenvalues of A_{BB} be $\theta_1 \geq \dots \geq \theta_{|B|} > 0$, and let U_q contain eigenvectors for the first q . Then

$$A_{BB} U_q = U_q \Theta_q, \quad U_q^* U_q = I_q, \quad \Theta_q = \text{diag}(\theta_1, \dots, \theta_q),$$

Following the EigenPro spectral preconditioner [20, 1], define

$$P_{\text{eig}}(B, q) = \begin{bmatrix} \theta_{q+1}^{-1} I_B - U_q (\theta_{q+1}^{-1} I_q - \Theta_q^{-1}) U_q^* & 0 \\ 0 & \Delta_T^{-1} \end{bmatrix}.$$

The analysis uses exact eigenpairs; the implementation substitutes the eigensolver outputs in the same formula.

In the exact case, Proposition 3.1 has $\mu_B^- = \theta_{|B|}/\theta_{q+1}$ and $\mu_B^+ = 1$. Thus the leading q eigenvalues on the box become one, while smaller eigenvalues are scaled uniformly. This correction is weaker than inversion on the same box, but its storage

and application costs grow linearly rather than quadratically in $|B|$. For $B = J_m$, this is $P_{\text{eig}}(J_m, q)$.

For Cartesian B , a product with A_{BB} costs $O(|B| \log |B|)$; details are given in Appendix B.1. If the eigensolver uses n_{eig} such products, they cost $O(n_{\text{eig}} |B| \log |B|)$, and orthogonalization costs $O(|B|q^2)$. Each application costs $O(|B|q)$. The rank q must therefore be large enough to save CG iterations but small enough that repeated applications remain inexpensive.

3.5. Algorithm and complexity. The applicable correction depends on box size. If a candidate box is small enough that its cubic construction and quadratic application costs are minor, the inverse is preferable because it removes the box contribution exactly. If an effective box is too large for inversion, $P_{\text{eig}}(B, q)$ permits the same box to be used at rank q . Increasing q strengthens the correction but also increases construction, storage, and every PCG iteration.

The endpoints are included in the same rule. When $B = J_m$, the inverse is a full inverse and the other endpoint is $P_{\text{eig}}(J_m, q)$. When B is a strict subset, both choices use the same diagonal scaling on T . Thus only two decisions are required: the box size and, for P_{eig} , the rank. The experiments compare a finite set of these choices by total time rather than by iteration count alone.

This comparison can be written directly in timing terms. Let t_A be the average time for one product with A , let t_P be the time for one preconditioner application, and let C_P be its construction time. If CG uses n_0 iterations and PCG uses n_P , then the preconditioner reduces iteration-related time when

$$C_P + n_P(t_A + t_P) < n_0 t_A,$$

when setup is shared by both solves. A configuration can therefore have the smallest iteration count and still lose in total time. Conversely, a moderate correction can be useful even when it leaves a larger condition number, provided that its construction and application are cheap.

PROPOSITION 3.7 (Validity for PCG). *If A is Hermitian positive definite and $0 \leq q < |B|$, then $P_{\text{inv}}(B)$ and $P_{\text{eig}}(B, q)$ are Hermitian positive definite. Hence both may be used in PCG.*

The result follows from the positive eigenvalues of the active operation and the positive diagonal on T ; a proof is given in Appendix A.6.

Algorithm 3.1 Construct the active-box preconditioner

Require: D, G, λ , budget s or threshold τ , and choice inv or eig

- 1: Compute $\rho_j = N|D_{jj}|^2/\lambda$ and select S .
 - 2: Let B be the smallest centered Cartesian box containing S , and set $T = J_m \setminus B$.
 - 3: Set $\Delta_T = \text{diag}(A_{TT})$.
 - 4: **if** choice is inv **then**
 - 5: Set $P_B = A_{BB}^{-1}$.
 - 6: **else**
 - 7: Estimate the leading $q + 1$ eigenvalues and the leading q eigenvectors of A_{BB} .
 - 8: Set P_B to the active block of $P_{\text{eig}}(B, q)$ defined in subsection 3.4.
 - 9: **end if**
 - 10: Return $P = \text{diag}(P_B, \Delta_T^{-1})$.
-

Let n_{eig} be the number of products with A_{BB} , n_{CG} the number of CG iterations,

and Q the number of prediction targets. Table 3.1 gives the full cost. Appendix B gives the PCG and end-to-end algorithms.

TABLE 3.1
Asymptotic costs. Preconditioner costs include diagonal scaling on T .

Operation	Build cost	Apply or solve cost	Memory
Fourier setup	$O(N + M \log M)$	–	$O(M)$
Product with A	–	$O(M \log M)$	$O(M)$
$P_{\text{inv}}(B)$	$O(M + B ^3)$	$O(M + B ^2)$	$O(M + B ^2)$
$P_{\text{eig}}(B, q)$	$O(M + n_{\text{eig}} B \log B + B q^2)$	$O(M + B q)$	$O(M + B q)$
CG with $P_{\text{inv}}(B)$	–	$O(n_{\text{CG}}(M \log M + M + B ^2))$	$O(M + B ^2)$
CG with $P_{\text{eig}}(B, q)$	–	$O(n_{\text{CG}}(M \log M + M + B q))$	$O(M + B q)$
Prediction at Q targets	–	$O(Q + M \log M)$	$O(M)$

4. Numerical Experiments. The experiments follow the comparison hierarchy dictated by the method’s scope. First, we compare directly with EFGP-CG, the reference line whose Fourier model our method accelerates; EFGP already reports broad comparisons with related large-scale KRR approximations [15]. We report the inverse and active-box EigenPro branches separately and count full-grid EigenPro as the limiting member of the latter. Second, we study box/rank and data-set variation under this same two-family organization, then repeat the comparison for squared exponential and Matérn kernels. Third, we briefly place the best member of each proposed family beside data-space Nyström and RPCholesky, EFGP+Jacobi, and general-purpose EigenPro baselines. Finally, we remove model-construction differences by giving every solver exactly the same hashed $A\beta = \mathbf{b}$.

The spectral methods in these comparisons form one proposed family rather than mutually unrelated algorithms. The active inverse and active-eigenpair routes spend a controlled budget on a score-selected Fourier box. The full-grid EigenPro route is the limiting member obtained when localization is removed and the spectral correction is constructed on the complete Fourier grid. Its stronger result is therefore evidence for the spectral preconditioning principle; the active default is the lower-setup, scalability-oriented member of the same family.

The end-to-end timing unit is each method’s own algorithmic setup plus solve. It excludes common data I/O, backend creation, host-to-device staging, and prediction, which is measured separately. We report all successful time–accuracy outcomes rather than suppressing an algorithm because its RMSE differs from another method by a fixed percentage. Fixed-system solver totals include active-set selection when applicable, preconditioner construction, and CG/PCG; their one shared Fourier construction is reported separately. Throughout this section, *EFGP-CG* means unpreconditioned CG in the reference EFGP implementation of Greengard et al. [15].

4.1. Setup. The synthetic task uses $x_i \sim \text{Unif}([0, 1]^2)$ and

$$f_{\star}(x) = \sin(2\pi x_1) + \frac{1}{2} \cos(2\pi x_2) + \frac{1}{5} \sin(2\pi(x_1 + x_2)).$$

Training targets are $y_i = f_{\star}(x_i) + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, 0.3^2)$. The four exact per-size artifacts use the same generation procedure and seeds (20260421 for training and 1 for testing), so their training arrays are consistent nested prefixes. Each file contains a noise-free test array of size $N/4$; formal reporting evaluates the first 2.5 million test rows. The retained original configuration sweep uses the same target function, noise level, and seed convention. Its timings are explicitly labelled as the original regime map and are never pooled with the later five-repeat replay or fixed-system medians.

The real-data task uses ground-class LiDAR points from the 2018 Winnebago USGS project [34, 35]. Horizontal coordinates are shifted and rescaled by a common factor and elevations are standardized using training-set statistics. Exact nested train artifacts are used at all four sizes, and formal evaluation uses the first 2.5 million rows of the fixed held-out test artifact.

The formal comparison uses the Matérn-3/2 kernel with lengthscale $\ell = 0.1$:

$$k_{\text{Matérn-3/2}}(x, z) = \left(1 + \frac{\sqrt{3}r}{\ell}\right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right),$$

where $r = \|x - z\|_2$. At Fourier-approximation tolerance 10^{-5} , the normalized support has $M = 35721$ modes [15, 6]. We use unit kernel variance, $\lambda = 0.1$, and relative residual tolerance 10^{-7} . The prospective broad-pipeline replay and fixed-system study run on an NVIDIA A100-SXM4-40GB GPU with one warm-up and five measured repetitions. The original family regime map is a retained configuration sweep and is labelled separately rather than pooled with those medians. Further settings and the older squared exponential setup are listed in Table C.3.

4.2. Primary comparison with EFGP-CG by proposed family. Our primary comparison does not compress the proposed method into a single automatic “default” row. For each data set and N , we report three quantities: EFGP-CG, the fastest tested active-inverse configuration, and the fastest tested active-box EigenPro configuration. The last family includes the full-grid limit $B = J_m$. Thus a full-grid winner is counted as a success of the proposed spectral family, not as an external baseline.

Table 4.1 gives the retained A100 Matérn sweep. The table shows time and test RMSE together; accuracy is not used as a binary filter for display. The two daggered inverse rows are the fastest recorded inverse attempts but were not selected by the original operation-level fit/convergence screen. We retain them to make the time-accuracy tradeoff visible and do not use them for a headline speed claim. The exact historical screen, the resulting regime map, and every original candidate row are reported in Appendix C.3.

TABLE 4.1

Primary A100 comparison with EFGP-CG, separated into the inverse and active-box EigenPro families. Times are complete recorded setup-plus-solve totals in seconds. Each proposed-family entry gives configuration, total, and test RMSE. J_m denotes the full Fourier grid. A dagger marks a raw inverse attempt rejected by the original screen; it is shown but not used for a speed claim.

Data	N	EFGP-CG		Inverse family			Active-box EigenPro family		
		Total	RMSE	Configuration	Total	RMSE	Configuration	Total	RMSE
Synthetic	10^7	4.553	0.008249	$ B = 2601$	0.553	0.008280	$ B = 5329, q = 256$	1.255	0.008281
	3×10^7	6.742	0.005238	$ B = 2601$	0.792	0.005262	$ B = 10609, q = 320$	1.600	0.005262
	10^8	10.539	0.002954	$ B = 5329$	1.757	0.002968	$J_m, q = 320$	2.044	0.002969
	3×10^8	15.328	0.001729	$ B = 5329$	2.862	0.001738	$J_m, q = 320$	3.345	0.001739
Winnebago	10^7	5.309	0.115140	$ B = 625$	4.550	0.115147	$J_m, q = 128$	0.979	0.115057
	3×10^7	8.643	0.113434	$ B = 625$	4.684	0.113583	$J_m, q = 128$	1.378	0.113378
	10^8	15.462	0.111901	$ B = 5329^\dagger$	6.984	0.150677	$J_m, q = 320$	1.603	0.111862
	3×10^8	27.748	0.110611	$ B = 5329^\dagger$	8.038	0.262925	$J_m, q = 384$	2.901	0.110583

The two branches are complementary. On Synthetic, the inverse family is fastest at all four displayed sizes and gives $5.36 \times - 8.52 \times$ complete-pipeline speedup over EFGP-CG. On Winnebago, the active-box EigenPro family is fastest and improves

from $5.42 \times$ at 10^7 to $9.56 \times$ at 3×10^8 . This is a data-dependent regime change, not an unexplained discontinuity in one fixed algorithm. It is precisely why we report the two families separately instead of allowing an automatic route switch to appear as one continuous “ours” curve.

4.3. Family-specific parameter and data-set variation. We next keep the two branches separate while changing their internal parameters and the data set. Candidate configurations are compared by total time, not by iteration count. Test RMSE is carried from the same row and is reported alongside time. This distinction matters because a larger inverse box or a higher EigenPro rank can shorten the solve while increasing the one-time construction cost.

TABLE 4.2

Within-family parameter variation for Synthetic Matérn at $N = 10^7$. Totals include the recorded binned setup, preconditioner construction, and PCG. The lowest iteration count is not necessarily the lowest total.

Family	Configuration	Iter.	Total (s)	Test RMSE
Inverse	$ B = 625$	594	0.976	0.0082805
	$ B = 961$	408	0.711	0.0082805
	$ B = 1369$	325	0.617	0.0082805
	$ B = 2601$	174	0.553	0.0082805
	$ B = 5329$	88	1.127	0.0082805
Active-box EigenPro	$ B = 2601, q = 192$	510	1.274	0.0082804
	$ B = 5329, q = 256$	416	1.255	0.0082805
	$ B = 10609, q = 320$	365	1.289	0.0082804
	$J_m, q = 256$	389	1.527	0.0082806

The inverse optimum is the intermediate box $|B| = 2601$: enlarging the box to 5329 halves the iteration count again but doubles construction sufficiently to make the total slower. The active-box EigenPro totals are flatter across the tested localized boxes, with $|B| = 5329, q = 256$ narrowly best. Hence the default-setting argument is family-specific; a single top- k /rank pair cannot stand in for both operations.

Data geometry can also reverse the family ordering. At $N = 3 \times 10^7$ with the same Matérn kernel, Synthetic favors the inverse branch (0.792 versus 1.600 seconds), whereas Winnebago favors the full-grid member of the active-box EigenPro branch (1.378 versus 4.684 seconds). The Fourier diagonal and score alone do not determine this switch: the sample-dependent Toeplitz term changes with the input distribution. The revised one-click artifact therefore writes a separate inverse row and a separate active-box/full-grid minimum for every parameter and data-set case; it never lets an automatic route switch masquerade as stability of one method.

4.4. Kernel comparison under the same two-family definition. The previous mechanism table mixed score fractions, eigenvalues, and a particular preconditioner without making the compared totals clear. We replace it with a direct kernel experiment. For each kernel and data set, Table 4.3 uses the family-optimal configuration selected by the procedure of subsection 4.2: one inverse result and the minimum over localized and full-grid EigenPro results. Every numeric column has the same meaning as in the primary table.

TABLE 4.3

Two-family comparison at $N = 3 \times 10^7$. Total is complete recorded setup plus solve in seconds; speedup is EFGP-CG total divided by the displayed family total. Full-grid EigenPro is denoted by J_m .

Data	Kernel	Family/configuration	Total	Test RMSE	Speedup	EFGP-CG
Synthetic	SE	Inverse, $B = J_m$	0.211	0.001071	14.70	3.098
	SE	Active-box EigenPro, $J_m, q = 128$	0.810	0.001073	3.82	3.098
	Matérn	Inverse, $ B = 2601$	0.792	0.005262	8.52	6.742
	Matérn	Active-box EigenPro, $ B = 10609, q = 320$	1.600	0.005262	4.21	6.742
Winnebago	SE	Inverse, $B = J_m$	0.212	0.374291	13.55	2.868
	SE	Active-box EigenPro, $J_m, q = 64$	0.499	0.374291	5.74	2.868
	Matérn	Inverse, $ B = 625$	4.684	0.113583	1.85	8.643
	Matérn	Active-box EigenPro, $J_m, q = 128$	1.378	0.113378	6.27	8.643

For SE, the grid has only 1225 modes and full inversion wins on both data sets. For Matérn, the grid has 35721 modes: localization makes the inverse branch best on Synthetic, while the global eigenpair correction is much better on Winnebago. The conclusion is therefore not that one preconditioner dominates universally, but that the proposed construction contains two useful computational regimes whose best member depends on kernel spectrum and data geometry.

4.5. Brief comparison with other complete KRR pipelines. We next report the other complete KRR pipelines as a secondary comparison. The table shows the largest scale, where scalability is most consequential; the complete 10^7 – 3×10^8 matrix is retained in the experiment artifact. No fixed relative-accuracy cutoff is used to remove a successful row: raw training total and test RMSE are reported together. The full-grid spectral row is the proposed family’s high-budget extreme, whereas Nyström and RPCholesky are data-space KRR algorithms and Jacobi is an EFGP baseline. EFGP-CG remains the primary reference and is already reported in Table 4.1.

TABLE 4.4

Secondary end-to-end KRR comparison at $N = 3 \times 10^8$. Training total is method-owned setup plus solve in seconds. “Resource limit” is the prospectively declared memory outcome, not a timed failure.

Method	Synthetic		Winnebago	
	Train total	Test RMSE	Train total	Test RMSE
Ours (binned default)	3.712	0.001740	2.682	0.090239
Full-grid spectral family member	6.442	0.001730	6.253	0.090214
EFGP+Jacobi	5.270	0.001730	31.406	0.090216
Nyström KRR	14.797	0.023004	14.793	0.275328
RPCholesky KRR	Resource limit		Resource limit	

At 3×10^8 , the binned default is the fastest successful route on both data sets. The full-grid family member is also successful: it preserves the EFGP-level RMSE while substantially improving on EFGP-CG, and is $1.74\times$ and $2.34\times$ slower than the cheaper binned default on Synthetic and Winnebago, respectively. Nyström exposes a different speed–accuracy tradeoff at smaller sizes; at the largest size shown here it is both slower and less accurate. RPCholesky completes the 10^7 cases, but its declared memory requirement exceeds the available budget from 3×10^7 onward, so we make no large- N timing claim for it.

General-purpose EigenPro line.. EigenPro2 and EigenPro3 attack the data-space kernel-training problem with GPU-oriented stochastic updates and Nyström spectral information [21, 1]. They therefore do not share the EFGP Fourier system or its residual-based stopping rule. We keep their comparison separate from the 10^7 – 3×10^8 complete-pipeline matrix and use time to an EFGP-CG validation target. The retained legacy EigenPro3 experiment uses 9×10^4 training, 10^4 validation, and 10^3 test points. For Matérn-3/2, the $p = 30000$ model does not reach the target within 30 epochs, whereas the proposed Fourier family reaches it in 0.50 seconds. For the Gaussian kernel, EigenPro3 reaches the target in 158.24 seconds versus 0.29 seconds for the proposed route. EigenPro2 was also considered as the predecessor in this line, but no matched, reproducible time-to-target row is retained in the formal artifact; consequently we do not turn its missing row into a numerical speedup or a convergence claim.

4.6. Controlled fixed-system comparison. To isolate the solver contribution, we select the Synthetic $N = 3 \times 10^7$ Matérn case and construct the Fourier weights, Toeplitz generator, regularization arrays, and right-hand side exactly once. All six methods receive the same zero initial iterate and immutable system, whose SHA-256 identifier begins 12c0a9811168. The common $A, \mathbf{b}$ construction takes 0.606 seconds and is excluded from every solver total. After one warm-up, five shuffled measured repetitions synchronize each timing boundary. For active methods the total includes fresh active-set selection, preconditioner construction, and solve; for CG it is the solve alone.

TABLE 4.5

Fixed-system A100 comparison at $N = 3 \times 10^7$. Components and totals are median-total representatives in seconds; S_{CG} is the median of matched-repeat CG/method total ratios. All five measured repetitions converge with independently recomputed relative residual below 10^{-7} .

Method	Iter.	Select	Build	Solve	Total	S_{CG}
CG	6185	0	0	6.525	6.525	1.000
Jacobi	1385	0	0.0005	1.761	1.761	3.715
Ours (frozen default)	495	0.0048	0.678	0.798	1.481	4.421
Active EigenPro	495	0.0048	0.714	0.809	1.527	4.272
Full-grid spectral family member	479	0	0.519	0.757	1.276	5.128
Active inverse	54	0.0049	3.734	0.146	3.884	1.667

The frozen default reduces the total from 6.525 to 1.481 seconds, a $4.42\times$ paired-median speedup over CG, and wins all five matched repetitions. Because A and $\mathbf{b}$ are identical, this confirms that the gain over the EFGP-CG line is not only a faster setup effect. More importantly, the full-grid family member reaches 1.276 seconds and a $5.13\times$ speedup. We interpret this as the high-budget limiting success of the proposed spectral family: removing localization spends less selection logic and captures the global difficult subspace more completely on this particular system. The default remains the deployment-oriented route because its localized construction is designed to scale when a full-grid spectral setup becomes less attractive. The active inverse illustrates the opposite tradeoff: its 54-iteration, 0.146-second solve is the shortest, but the 3.734-second factor construction raises its total to 3.884 seconds.

5. Discussion and Limitations. The present scope is low-dimensional KRR with stationary kernels. The Fourier grid grows exponentially with dimension, and a nonstationary kernel does not give the same Toeplitz structure. A centered box is also

best aligned with an isotropic spectrum; anisotropic or irregular spectra may require a different active set.

Active-box effectiveness depends on spectral alignment and coupling. The score ρ_j uses the Fourier weights, N , and λ , but it does not see the sample-dependent off-diagonal entries of G . A useful box must therefore contain the difficult directions while leaving a complement with small diagonal error and weak coupling. The terms δ_T and γ_{BT} in Proposition 3.1 retain these effects.

The end-to-end evidence is strongest relative to EFGP-CG, the model line that the proposed pipeline directly accelerates. Across 10^7 – 3×10^8 samples, Synthetic favors the inverse family (5.36 – $8.52\times$ speedup), while Winnebago favors the active-box EigenPro family (5.42 – $9.65\times$). Reporting those branches separately avoids interpreting a data-dependent route switch as a sudden scaling effect. The secondary KRR comparison is deliberately more modest: Nyström exposes a visible speed–accuracy tradeoff at smaller scales, and RPCholesky reaches its declared memory limit above 10^7 samples. These outcomes are reported rather than removed by a fixed relative-accuracy gate.

Preconditioning cannot remove the $O(N)$ Fourier setup cost. It reduces the number of products after the system is built, but the type-1 NUFFTs for the Toeplitz generator and right-hand side remain. Once setup dominates, further reductions in the iteration count have limited effect on total time.

The two empirical levels should nevertheless not be merged. End-to-end KRR timing includes each method’s own algorithmic setup and solve, whereas the controlled comparison gives all solvers one identical Fourier system. In the latter experiment the frozen default is $4.42\times$ faster than CG, so the gain is not solely a setup effect. The full-grid spectral member reaches a $5.13\times$ speedup and is the fastest family member on that frozen system. This is the high-budget extreme of the proposed construction rather than an external algorithm defeating it. Localization should therefore be viewed as a data-dependent construction/application tradeoff: the default limits setup and storage, while the full-grid member can be stronger when the difficult subspace is global. The active inverse is slower in total despite its short solve because factor construction dominates.

The reported end-to-end total is method-owned algorithmic training time, not process wall clock. Common data I/O, backend creation, and host-to-device staging are excluded, and prediction is separate. The fixed-system total similarly excludes the one 0.606-second shared construction of A and $\mathbf{b}$. These accounting choices expose the setup and solver effects but should be preserved when reproducing or comparing the numbers.

6. Conclusion. A fixed Fourier-discretized KRR system can be severely ill-conditioned even when each matrix-vector product is fast. The proposed active inverse and leading-eigenpair corrections reduce the solve cost when difficult directions are concentrated in a Fourier box and its coupling to the complement is small. Combined with the binned setup, the inverse and eigenpair branches give up to $8.52\times$ and $9.65\times$ complete-pipeline speedup over EFGP-CG, respectively, while their time–accuracy outcomes remain explicit. The identical- $A, \mathbf{b}$ experiment confirms a $4.42\times$ solver-total gain for the default and a $5.13\times$ gain for the full-grid family member. Comparisons with Jacobi, Nyström, RPCholesky, and the different spectral-family budgets delimit the claim: the family substantially accelerates the EFGP-CG line, while the preferred member depends on the cost of localization, factorization, and global eigenspace construction.

7. Code and Reproducibility. Code and experiment scripts are available at <https://github.com/Yifiwif/EFGP-Eigenpro>. The repository records the software dependencies, random seeds, data splits, solver tolerances, timing scope, and tested configurations. The formal one-click output bundle stores repeat-level rows together with reporter-generated summaries, so the paper tables can be regenerated without repeating the 300-million-sample runs. The five-repeat broad-comparison and fixed-system values in this paper use the output bundle `paper_one_click_0034442c4bc1`, generated from repository commit `0034442c4bc1a130880dc08241b04a21ead17c56`.

The two-family values in Tables 4.1 to 4.3 come from the tracked original candidate/selected tables identified in Appendix C.3. The revised one-click notebook exports those rows as `archived_original_full_results.csv` and `archived_regime_map_two_family.csv`. The secondary end-to-end values in Table 4.4 come from `stage1_end_to_end_krr/stage1_scale_summary.csv`. The controlled values in Table 4.5 come from `two_stage_formal_report/stage2_formal_solver_totals.csv`, with the shared construction recorded in `two_stage_formal_report/stage2_shared_ab_setup.csv`. The corresponding case directory contains the immutable system manifest, matched warm-up and measured rows, component hashes, and independently recomputed residuals. Its system identifier begins `12c0a9811168`; the manifest stores the full SHA-256 value.

Row-level metadata record the setup route, effective box size and rank, convergence, residual, timing components, prediction metrics, and explicit resource-limit outcomes. Full numerical settings, candidate grids, hardware, and timing accounting are given in Appendix C. Older controlled outputs and acquisition pilots remain in the repository as supplementary provenance but do not supply the formal tables above.

Appendix A. Proofs.

A.1. Proof of the threshold-box growth result.

Proof. By (2.3), the threshold condition is

$$(A.1) \quad \widehat{k}(hj) > \frac{\tau\lambda}{Nh^d}.$$

For the squared exponential kernel,

$$\widehat{k}_{SE}(\xi) = (\sqrt{2\pi}\ell)^d \exp(-2\pi^2\ell^2\|\xi\|_2^2).$$

Substitution in (A.1) gives

$$2\pi^2\ell^2h^2\|j\|_2^2 < \log\left(\frac{Nh^d(\sqrt{2\pi}\ell)^d}{\tau\lambda}\right).$$

If the logarithm is nonpositive, the strict inequality has no solution; otherwise it is equivalent to (2.6) and (2.7).

For the Matérn kernel, (2.8) turns (A.1) into

$$2\nu + 4\pi^2\ell^2h^2\|j\|_2^2 < \left(\frac{Nh^d c_{\nu,d,\ell}}{\tau\lambda}\right)^{1/(\nu+d/2)}.$$

Using $1/(\nu+d/2) = 2/(2\nu+d)$, and again observing that a nonpositive squared radius gives an empty strict-threshold set, proves (2.9) and (2.10).

Both nonempty threshold sets are discrete Euclidean balls intersected with J_m . Every selected index therefore satisfies $\|j\|_\infty \leq \|j\|_2 < R_\tau$. Since b_τ is an integer,

$$b_\tau \leq \min\{m, \lceil R_\tau \rceil - 1\},$$

which proves (2.11). If $R_\tau > m$, every axis index $\pm me_s$, $s = 1, \dots, d$, is selected, so the smallest centered box is J_m . The fixed-grid growth laws follow directly from the two exact radii before this truncation occurs. $\square$

A.2. Proof of the active-box spectral bound.

Proof. Write

$$P^{1/2} A P^{1/2} = \begin{bmatrix} P_B^{1/2} A_{BB} P_B^{1/2} & P_B^{1/2} A_{BT} \Delta_T^{-1/2} \\ \Delta_T^{-1/2} A_{TB} P_B^{1/2} & \Delta_T^{-1/2} A_{TT} \Delta_T^{-1/2} \end{bmatrix}.$$

The eigenvalues of its block-diagonal part lie in $[\min(\mu_B^-, 1 - \delta_T), \max(\mu_B^+, 1 + \delta_T)]$. The remaining Hermitian matrix is

$$\begin{bmatrix} 0 & P_B^{1/2} A_{BT} \Delta_T^{-1/2} \\ \Delta_T^{-1/2} A_{TB} P_B^{1/2} & 0 \end{bmatrix},$$

and its spectral norm is γ_{BT} . Weyl's inequality gives the first line of (3.1). If the stated lower endpoint is positive, dividing the upper eigenvalue bound by the lower one gives the second line. $\square$

A.3. Proof of the first Fourier-system bound. Since $G_{j,j} = N$, we have $\Delta_T = N D_T^2 + \lambda I_T$. Hence

$$\begin{aligned} \Delta_T^{-1/2} A_{TT} \Delta_T^{-1/2} - I_T &= \Delta_T^{-1/2} D_T (G_{TT} - N I_T) D_T \Delta_T^{-1/2} \\ &= W_T \left(\frac{G_{TT}}{N} - I_T \right) W_T, \end{aligned}$$

where $W_T = \sqrt{N} \Delta_T^{-1/2} D_T$. Its diagonal entries satisfy $(W_T)_{j,j}^2 = \rho_j / (1 + \rho_j)$. If $B \supseteq S_\tau$, then $\rho_j \leq \tau$ on T , and therefore

$$\delta_T \leq \|W_T\|_2^2 \left\| \frac{G_{TT}}{N} - I_T \right\|_2 \leq \frac{\tau}{1 + \tau} \left\| \frac{G_{TT}}{N} - I_T \right\|_2.$$

This is the first bound in (3.2).

A.4. Proof of the second Fourier-system bound. The ridge term has no off-diagonal block, so $A_{BT} = D_B G_{BT} D_T$. Thus

$$\begin{aligned} \gamma_{BT} &= \left\| P_B^{1/2} D_B G_{BT} D_T \Delta_T^{-1/2} \right\|_2 \\ &\leq \left\| \sqrt{N} P_B^{1/2} D_B \right\|_2 \left\| \frac{G_{BT}}{N} \right\|_2 \left\| \sqrt{N} D_T \Delta_T^{-1/2} \right\|_2. \end{aligned}$$

The last factor is at most $\sqrt{\tau / (1 + \tau)}$ by the preceding calculation. This is the second bound in (3.2).

A.5. Proof of the population Fourier Gram results.

Proof. The definition of G gives

$$\frac{G_{j,\ell}}{N} = \frac{1}{N} \sum_{n=1}^N e^{2\pi i h \langle \ell - j, x_n \rangle}.$$

Taking expectations proves $H_{j,\ell} = \psi(\ell - j)$. For fixed r , let $Z_n(r) = e^{2\pi i h \langle r, x_n \rangle}$. Independence, $|Z_n(r)| = 1$, and $\mathbb{E}Z_n(r) = \psi(r)$ give

$$\mathbb{E} \left| \frac{1}{N} \sum_{n=1}^N Z_n(r) - \psi(r) \right|^2 = \frac{1}{N} \mathbb{E} |Z_1(r) - \psi(r)|^2 = \frac{1 - |\psi(r)|^2}{N}.$$

For a Gaussian input, its characteristic function gives (3.5). The covariance lower bound implies

$$|\psi(r)| \leq e^{-a\|r\|_2^2}, \quad a = 2\pi^2 h^2 \sigma_x^2.$$

The diagonal of H is one. For each $j \in T$, distinct indices obey

$$\sum_{\substack{\ell \in T \\ \ell \neq j}} |H_{j,\ell}| \leq \sum_{r \in \mathbb{Z}^d \setminus \{0\}} e^{-a\|r\|_2^2} = \eta_d(a).$$

The Hermitian row-sum bound proves the first inequality in (3.7). Every row sum and every column sum of the rectangular block H_{BT} is bounded by the same punctured lattice sum, so

$$\|H_{BT}\|_2 \leq \sqrt{\|H_{BT}\|_1 \|H_{BT}\|_\infty} \leq \eta_d(a).$$

The lattice sum factorizes as

$$1 + \eta_d(a) = \left(1 + 2 \sum_{q=1}^{\infty} e^{-aq^2} \right)^d.$$

Because e^{-ax^2} decreases on $[1, \infty)$,

$$\sum_{q=1}^{\infty} e^{-aq^2} \leq e^{-a} + \int_1^{\infty} e^{-ax^2} dx \leq e^{-a} + \int_1^{\infty} x e^{-ax^2} dx = e^{-a} \left(1 + \frac{1}{2a} \right).$$

This proves (3.8).

The strong law applied to the real and imaginary parts of each generator coefficient gives $g_r/N \rightarrow \psi(r)$ almost surely. The set J_{2m} is finite, so all coefficients converge simultaneously almost surely. Entrywise convergence then implies (3.9) because the matrix dimension is fixed. Integrating the characteristic exponential over the product-uniform density on $[0, 1]^d$ also gives (3.10). $\square$

Proof of Corollary 3.5. The triangle inequality gives

$$\left\| \frac{G_{TT}}{N} - I_T \right\|_2 \leq \varepsilon_{TT} + \|H_{TT} - I_T\|_2, \quad \left\| \frac{G_{BT}}{N} \right\|_2 \leq \varepsilon_{BT} + \|H_{BT}\|_2.$$

Substitution into (3.2) proves (3.12) and (3.13). The Gaussian specialization follows from (3.7), and the convergence statement follows by restriction of (3.9) to fixed blocks. $\square$

A.6. Proof of Proposition 3.7.

Proof. Every principal submatrix of a Hermitian positive-definite matrix is Hermitian positive definite. Thus A_{BB}^{-1} is positive definite. Moreover, $\Delta_T = \text{diag}(A_{TT})$ has strictly positive entries, so Δ_T^{-1} is positive definite. Their block diagonal combination is therefore positive definite, which proves the claim for $P_{\text{inv}}(B)$.

For $P_{\text{eig}}(B, q)$, complete U_q to a unitary eigenbasis of A_{BB} . On the direction u_i with $i \leq q$, the active block of $P_{\text{eig}}(B, q)$ has eigenvalue θ_i^{-1} . On every orthogonal direction it has eigenvalue θ_{q+1}^{-1} . All these values are positive, and the complementary block Δ_T^{-1} is also positive definite. Hence $P_{\text{eig}}(B, q)$ is Hermitian positive definite. $\square$

Appendix B. Algorithms and Implementation Details.

This appendix gives the standard PCG iteration and the complete Fourier KRR solver. The main text defines the preconditioners and summarizes their costs.

Algorithm B.1 Preconditioned conjugate gradient

```

1: Choose  $\beta_0$ . Set  $r_0 = b - A\beta_0$ ,  $z_0 = Pr_0$ , and  $p_0 = z_0$ .
2: for  $t = 0, 1, 2, \dots$  do
3:    $w_t = Ap_t$ 
4:    $\alpha_t = (r_t^* z_t) / (p_t^* w_t)$ 
5:    $\beta_{t+1} = \beta_t + \alpha_t p_t$ 
6:    $r_{t+1} = r_t - \alpha_t w_t$ 
7:   if  $\|r_{t+1}\|_2 / \|b\|_2 \leq \varepsilon_{\text{CG}}$  then
8:     break
9:   end if
10:   $z_{t+1} = Pr_{t+1}$ 
11:   $\eta_t = (r_{t+1}^* z_{t+1}) / (r_t^* z_t)$ 
12:   $p_{t+1} = z_{t+1} + \eta_t p_t$ 
13: end for
```

B.1. Products on the active box. The Toeplitz generator defined in subsection 2.3 determines the block through $(G_{BB})_{j,\ell} = g_{\ell-j}$ for $j, \ell \in B$. Thus $A_{BB} = D_B G_{BB} D_B + \lambda I_B$ is obtained without forming A or revisiting the data. The reported inverse runs form A_{BB}^{-1} from a Cholesky factorization and two triangular solves, then store the dense inverse. For $P_{\text{eig}}(B, q)$, products with A_{BB} use the same Toeplitz structure as products with A . The reported runs compute the largest $q+1$ eigenpairs in complex double precision with GPU `eigsh`, using tolerance 10^{-3} and at most $5q$ iterations. The recorded construction time includes active-box construction and the inverse build or eigensolve.

B.2. Optional matrix-free spectral diagnostics. For a chosen strict box, the two global terms in Proposition 3.1 can be estimated without forming A . Define

$$R_T = \Delta_T^{-1/2} A_{TT} \Delta_T^{-1/2} - I_T, \quad \delta_T = \|R_T\|_2.$$

Applying R_T requires one product with A , restriction to T , and diagonal scaling. Because R_T is Hermitian but need not be positive semidefinite, a Lanczos estimate of δ_T must approximate the largest-magnitude Ritz value, equivalently both spectral endpoints.

The coupling can be estimated without constructing $P_B^{1/2}$, since

$$(B.1) \quad \gamma_{BT}^2 = \lambda_{\max} \left(\Delta_T^{-1/2} A_{TB} P_B A_{BT} \Delta_T^{-1/2} \right).$$

This operator is positive semidefinite; each application follows the sequence $T \rightarrow B \rightarrow B \rightarrow T$ using A_{BT} , P_B , and A_{TB} . Hermitian Lanczos iterations can therefore provide both estimates using existing matrix-vector products [18, 19]. For the inverse correction $\mu_B^- = 1$; for $P_{\text{eig}}(B, q)$ with exact eigenpairs, $\mu_B^- = \theta_{|B|}/\theta_{q+1}$. Together the estimates give the lower endpoint

$$L_B = \min\{\mu_B^-, 1 - \delta_T\} - \gamma_{BT}.$$

A nonpositive L_B means that the condition-number bound in (3.1) is vacuous, not that PCG must fail. These optional diagnostics are not needed to construct the preconditioner and should be run outside any timed solve.

B.3. Complete solver. The complete algorithm joins Fourier setup, preconditioner construction, PCG, and prediction.

Algorithm B.2 Fourier KRR with active-box preconditioning

Require: Data $\{(x_n, y_n)\}_{n=1}^N$, kernel k , regularization λ , Fourier parameters h, m , Fourier setup route, budget s or threshold τ , choice inv or eig, tolerance ε_{CG} , and maximum iteration count t_{max}

Ensure: Fourier coefficients β and prediction function $\hat{f}$

- 1: Using the chosen setup route, construct J_m , D , the Toeplitz generator g , and $b = DF^*\mathbf{y}$. The routes used in the tables are recorded in Appendix C.
 - 2: Apply A by $u \mapsto DG(Du) + \lambda u$.
 - 3: Construct $P_{\text{inv}}(B)$ or $P_{\text{eig}}(B, q)$ using Algorithm 3.1.
 - 4: Solve $A\beta = b$ by Algorithm B.1, stopping at t_{max} if the tolerance is not reached.
 - 5: Evaluate $\hat{f}(x) = \sum_{j \in J_m} \beta_j D_{jj} e^{2\pi i h \langle j, x \rangle}$ at target points using a type-2 NUFFT.
-

Appendix C. Experiment Configuration Details.

This appendix separates the end-to-end KRR protocol from the protocol used to isolate the Fourier linear solver. The saved bundle contains the formal six-pipeline broad-comparison matrix over 10^7 – 3×10^8 , explicit inverse/eigenpair family replays, the earlier EFGP-only regime map, and the formal matched fixed-system rows. Each row records the data set, kernel, N , effective configuration, residual, timing components, and prediction metrics. Medians and paired ratios are formed only within one protocol and never combine timings from different systems.

C.1. Stage protocols. Table C.1 separates the accounting rules. Stage 1 compares KRR algorithms, so each method pays for its own model construction. Stage 2 compares solvers, so every method receives exactly the same A and $\mathbf{b}$.

TABLE C.1
Roles and accounting of the two formal experimental stages.

Quantity	Stage 1: end-to-end KRR	Stage 2: fixed $A\mathbf{\beta} = \mathbf{b}$
Primary question	Accuracy and setup-plus-solve time of complete KRR pipelines from 10^7 to 3×10^8 samples	Solver acceleration on one identical Fourier system per case
Methods	Primary family replay: EFGP-CG, explicit inverse, localized EigenPro, and full-grid EigenPro; secondary broad matrix: Nyström KRR, RP-Cholesky KRR, EFGP+Jacobi, and the corresponding EFGP rows	CG; Jacobi-PCG; proposed full-grid, active-inverse, and active-eigenpair PCG
System construction	Each KRR method performs and pays for its own setup	One shared, hashed $D, G, \mathbf{b}, \lambda$ for all methods
Timing unit	Method-specific setup plus solve; prediction reported separately	Preconditioner selection/build plus solve; shared Fourier setup separate
Selection	Inverse and eigenpair configurations fixed separately before timing; all successful raw time and accuracy rows retained	Configuration frozen before matched timing
Repetition	One warm-up plus five measured repetitions on the common A100	One warm-up plus five shuffled, paired measured repetitions
Arithmetic and check	Per-method residual or optimization check and test RMSE	fp64/complex128 and independently recomputed true residual
Numerical status	Complete matrix with explicit resource-limit outcomes	Complete matched A100 run with verified common system hash

The formal fixed-system target is the Synthetic $N = 3 \times 10^7$ Matérn case selected by the completed scale campaign. Its shared construction cannot be added to or substituted for any end-to-end KRR row. Likewise, the earlier regime-map ratios remain a separate historical protocol.

C.2. Formal Stage 1 end-to-end KRR campaign. The primary family scale profile uses `efgp-standard-cg`, `efgp-standard-full-eig`, `ours-binned-inverse`, and `ours-binned-active-eig`. The inverse and active-eigenpair routes carry separate transferred top- k /box/rank fields. The reporter keeps the inverse row and minimizes current median total over localized and full-grid EigenPro for the second family. It never minimizes across the two families. The kernel profile applies the same rule separately to SE and Matérn, and the robustness profile repeats it at every parameter or data-set value.

The secondary broad scale matrix uses $N \in \{10^7, 3 \times 10^7, 10^8, 3 \times 10^8\}$ on the same declared training and test artifacts. Plain EFGP-CG and EFGP+Jacobi run as reference EFGP pipelines. The proposed family includes the full-grid spectral route and the first-order binned default followed by its frozen active inverse/eigenpair rule. Nyström KRR and RPCholesky KRR act on the data-space kernel problem; their landmark or pivot selection, feature/factor construction, and solve are part of setup and solve time. A Fourier-space Nyström or RPCholesky preconditioner is not a replacement for either data-space row.

For every method, the primary training total is $T_{\text{setup}} + T_{\text{solve}}$. These two components are also reported separately so that acceleration from faster setup is not attributed to the solver, or vice versa. This is the method-owned algorithmic training total rather than process wall clock: common data-file input, backend creation, and host-to-device staging are excluded by declaration. Prediction time is reported separately. All successful time-accuracy outcomes remain visible; reference-relative

accuracy labels are descriptive metadata and do not remove raw timings. All methods use the same hardware and repeated-timing policy. Unavailable, resource-limited, or timed-out rows remain explicit rather than disappearing from a curve.

Exact simple RPCholesky retains its rank-by- N factor. At the frozen rank 256 in fp64 this factor exceeds the declared 48-GiB cap from $N = 3 \times 10^7$ onward, so a `resource_limit` row is the expected scalability outcome, not a pilot-method substitution. It counts as a complete protocol outcome but not as a timed speed comparison. The completed scale study then freezes the $N = 3 \times 10^7$ Synthetic target before Stage 2 timing.

C.3. Archived regime map and original full-result audit. The archived experiment is retained for two purposes: it records the broad regime map used to choose configurations before the prospective replay, and it exposes every original candidate rather than only the row printed in the main text. The authoritative Matérn files are `diagnostic_tables/paper_table1_candidates.csv` and `diagnostic_tables/paper_table1_selected.csv` inside the tracked `btabs_group_a_group_b_group_c_20260703_053504` export bundle. The SE rows use the later corrective `btabs_group_b_20260704_081848` bundle. The one-click notebook copies these into the run directory as `archived_original_full_results.csv` and `archived_original_operation_selected.csv`, and also writes the collapsed `archived_regime_map_two_family.csv`. Thus the complete row-level result is part of the saved paper artifact even though it is too large to typeset in full.

Table C.2 states the selection procedure. It is important to separate its historical screen from the current reporting policy. The old sweep used a training-fit screen to prevent a nonconverged candidate from becoming the representative timing row. The present paper does not use that screen to hide a successful time–accuracy tradeoff: raw attempts and test RMSE remain in the full-result CSV, and rejected inverse attempts are marked explicitly when displayed. The old rule is reported because it explains exactly how the transferred configurations were obtained.

TABLE C.2

Auditable hierarchy from original candidates to the two-family tables. No current-run timing is consulted when configurations are transferred.

Step		Rule
Candidate	uni-	Keep every recorded EFGP-CG, active-inverse, localized Box-EigenPro, and full-grid EigenPro-PCG row, including failed or capped attempts.
Historical	opera-	
screen		Within each data set, kernel, and N , require the archived convergence status and training RMSE no larger than 1.10 times EFGP-CG. Test RMSE is not used for selection.
Operation	winner	Among rows passing that historical screen, minimize recorded complete pipeline total separately for Active inverse, Box-EigenPro, and full-grid EigenPro-PCG.
Family	collapse	The inverse operation is the inverse family. Box-EigenPro and full-grid EigenPro-PCG are one active-box EigenPro family; choose the smaller operation winner by total time.
Main-text	display	Report EFGP-CG, the inverse-family result, and the active-box EigenPro-family result together with test RMSE. Any shown raw row outside the old screen is marked and excluded from the headline speed claim rather than deleted.
Prospective	re-	Transfer only top- k , observed box size, and rank. Rerun one warm-up plus five measured repetitions; select the faster localized/full-grid eigenpair row from the current medians. Archived time never substitutes for replay time.
play		

The archived regime map is interpretable only at the family level. For SE, $M = 1225$ makes full inversion the usual winner. For Matérn, Synthetic selects moderate inverse boxes, whereas Winnebago increasingly selects the global eigenpair limit as N grows. These transitions generate the frozen family-specific configurations used by the revised `family_scale_10m_300m`, `family_robustness_at_selected_target`, and `family_kernel_at_30m` notebook profiles. The original full candidate CSV remains the source for verifying every alternative box, rank, iteration count, timing component, and accuracy value.

C.4. Legacy EFGP scale-map settings. Table C.3 collects the common parameters and the two Fourier setup routes. The six training sizes are used for both the synthetic and USGS tasks. The kernels differ in Fourier dimension because the same Fourier approximation tolerance is applied to different spectral densities.

TABLE C.3
Settings used in the archived exploratory scale map.

Quantity	Value
Training sizes N	$10^6, 3 \times 10^6, 10^7, 3 \times 10^7, 10^8, 3 \times 10^8$
Kernel variance	1
Lengthscale ℓ	0.1
Regularization λ	0.1
Fourier approximation tolerance	10^{-5}
NUFFT tolerance	10^{-10}
Fourier setup routes	direct (baseline); first-order binned (candidates)
Relative residual tolerance	10^{-7}
Maximum CG iterations	80000 (baseline), 3000 (preconditioned)
Fourier dimensions	1225 (SE), 35721 (Matérn-3/2)
Device	NVIDIA A100-SXM4-40GB

The relative residual is $\|b - A\beta_t\|_2 / \|b\|_2$. A run is marked converged only when this value reaches the tolerance before the iteration limit. For the synthetic task, the data seeds are 20260421 (training) and 1 (test), and the noise-free test set has $N/4$ points. The USGS test set is the fixed held-out portion described in subsection 4.1. All methods within a row use the same training and test data. Synthetic training targets use Gaussian noise with standard deviation 0.3.

C.5. Candidate boxes and ranks. The requested score budget s is not generally equal to $|B|$: the smallest centered Cartesian box may include additional indices. The actual box sizes, rather than s , determine memory and runtime. The values represented in the saved main sweep are summarized in Table C.4. Not every listed box–rank combination is run for every N ; the complete Cartesian sweep over the final three Matérn boxes and five ranks is the one reported in Table C.7.

TABLE C.4

Candidate score budgets, resulting box sizes, and EigenPro ranks in the saved experiment bundle. Entries in braces are aligned by position.

Kernel	Operation	Requested s / observed $ B $	Candidate q
SE	inverse	{512, 1024}/{625, 1225}	—
SE	EigenPro	{256, 512, 768}/{361, 625, 961}	{64, 128}
Matérn	inverse	{512, 728, 1024, 2048, 4096, 8192, 12288}	—
		{625, 961, 1369, 2601, 5329, 10609, 15625}	
Matérn	EigenPro	{2048, 4096, 8192, 16384}/{2601, 5329, 10609, 21025}	{192, 256, 320}
		{20480, 25720, 35721}/{25921, 32761, 35721}	{192, 256, 320, 384, 448}

These sets explain two conventions used in the main text. First, a configuration is always labeled by the observed $|B|$, because this is the dimension on which the operation is performed. Second, $B = J_m$ is written as $P_{\text{inv}}(J_m)$ or $P_{\text{eig}}(J_m, q)$, without assigning a new method name.

C.6. Legacy active-box alternatives on the largest systems. Table C.5 reports the alternatives omitted from the main scalability tables. A full-grid configuration is written with $B = J_m$; it is not assigned another method name.

TABLE C.5

Largest Matérn-3/2 systems ($M = 35721$). “No candidate met the screen” means that no tested inverse converged within 3000 iterations and passed the training-RMSE check. Speedup compares the recorded complete pipelines.

Data	N	Method	Configuration	Iter.	T_{tot}	Test RMSE	Speedup
Synthetic	10^8	EFGP-CG	—	8831	10.54	2.95×10^{-3}	1.00
Synthetic	10^8	$P_{\text{inv}}(B)$	$ B = 5329$	187	1.76	2.97×10^{-3}	6.00
Synthetic	10^8	$P_{\text{eig}}(B, q)$	$ B = 35721, q = 320$	495	2.04	2.97×10^{-3}	5.16
Synthetic	3×10^8	EFGP-CG	—	11494	15.33	1.73×10^{-3}	1.00
Synthetic	3×10^8	$P_{\text{inv}}(B)$	$ B = 5329$	246	2.86	1.74×10^{-3}	5.36
Synthetic	3×10^8	$P_{\text{eig}}(B, q)$	$ B = 35721, q = 320$	643	3.34	1.74×10^{-3}	4.58
USGS	10^8	EFGP-CG	—	13539	15.46	1.12×10^{-1}	1.00
USGS	10^8	$P_{\text{inv}}(B)$	no candidate met the screen	—	—	—	—
USGS	10^8	$P_{\text{eig}}(B, q)$	$ B = 35721, q = 320$	264	1.60	1.12×10^{-1}	9.65
USGS	3×10^8	EFGP-CG	—	22466	27.75	1.11×10^{-1}	1.00
USGS	3×10^8	$P_{\text{inv}}(B)$	no candidate met the screen	—	—	—	—
USGS	3×10^8	$P_{\text{eig}}(B, q)$	$ B = 35721, q = 384$	320	2.90	1.11×10^{-1}	9.56

C.7. Row selection and timing. For each data set, kernel, and N , EFGP-CG supplies the baseline training RMSE. A preconditioned row is eligible if it converges and its training RMSE is at most 1.10 times that baseline. Among eligible rows of a given operation, the row with the smallest total time is reported. Test RMSE is copied from that same row; it is not used to choose between candidates. When no row passes both conditions, the table states that no candidate met the screen.

Four timing fields are distinguished. Fourier setup includes the spectral weights, Toeplitz generator, and right-hand side. The baseline rows use direct cuFINUFFT setup; the preconditioned rows use the first-order binned route. The saved label for the latter is `binned_c1`. The two routes target the same prescribed Fourier modes but do not hold the assembled matrix and right-hand side fixed. Preconditioner construction starts after the setup recorded for its row. PCG time in Table C.6 excludes construction, while total time includes setup, construction, and PCG. Prediction is measured separately and is not included in total time. Consequently, the reported speedup compares complete recorded pipelines. The matched reruns in subsection 4.6 provide the separate fixed-system solver comparison.

TABLE C.6

Timing breakdown in seconds for the fastest configuration in each largest Matérn case. Binned denotes the first-order binned setup route. The PCG column excludes preconditioner construction.

Data	N	Setup route	Method	Setup	Prec. build	PCG	Total
Synthetic	10^8	Direct	EFGP-CG	1.00	–	9.54	10.54
Synthetic	10^8	Binned	$P_{\text{inv}}(B)$	0.53	0.88	0.35	1.76
Synthetic	3×10^8	Direct	EFGP-CG	2.99	–	12.34	15.33
Synthetic	3×10^8	Binned	$P_{\text{inv}}(B)$	1.53	0.88	0.46	2.86
USGS	10^8	Direct	EFGP-CG	1.02	–	14.44	15.46
USGS	10^8	Binned	$P_{\text{eig}}(J_m, 320)$	0.51	0.60	0.49	1.60
USGS	3×10^8	Direct	EFGP-CG	3.43	–	24.32	27.75
USGS	3×10^8	Binned	$P_{\text{eig}}(J_m, 384)$	1.55	0.74	0.61	2.90

The output bundle also retains the unrounded values used to compute speedup. Displayed times are rounded to two decimals, so recomputing a ratio from the printed values can differ slightly from the printed speedup. Failed runs remain in the raw tables and are excluded only by the stated convergence and fit rules.

C.8. Legacy EFGP cost breakdown. The alternative configurations for the largest Matérn systems are reported once, in Table C.5. On USGS, no tested inverse passes the screen at $N = 10^8$ or 3×10^8 , whereas $P_{\text{eig}}(J_m, q)$ gives $9.65\times$ and $9.56\times$ complete-pipeline speedups.

Iteration counts alone do not include the cost of constructing a preconditioner. Table C.6 separates the largest Matérn runs into Fourier setup, preconditioner construction, and PCG after construction. It also records the two setup routes, so the faster binned setup is not attributed to preconditioning. The components sum to total time up to rounding.

For the selected inverse, construction is more expensive than the subsequent PCG solve, but their sum is much smaller than the solve time of the recorded baseline. The same statement holds for the EigenPro rows. At $N = 3 \times 10^8$, the binned setup is about 54% of each selected total. Even if construction and PCG took no time, leaving that setup unchanged could improve the selected totals by only about $1.9\times$. Further gains must therefore reduce setup as well as iterations. The table also shows why candidates are selected by total time rather than iteration count. Dividing the solve times by their iteration counts shows that PCG costs about 1.7–1.8 times as much per step, but the saved iteration reduction is much larger than the one-time build cost expressed in baseline steps.

C.9. Legacy active-box size and EigenPro rank. Table C.7 reports all 15 configurations in the detailed USGS Matérn sweep at $N = 3 \times 10^8$. The actual box size is reported after enclosing the selected Fourier indices; it may exceed the requested top- s budget. All converged rows have test RMSE between 0.11058 and 0.11059, so this comparison is determined by computation rather than prediction error.

TABLE C.7

Total time in seconds / PCG iterations for $P_{\text{eig}}(B, q)$ on USGS with $N = 3 \times 10^8$ and $M = 35721$. The dagger marks a run that did not reach tolerance within 3000 iterations.

$ B \backslash q$	192	256	320	384	448
25921	7.05/3000 [†]	8.92/3000 [†]	6.99/2783	6.36/2232	5.95/1905
32761	7.12/3000 [†]	7.53/3000 [†]	6.81/2557	5.92/1946	5.64/1639
35721	3.29/794	3.01/541	2.91/408	2.90/320	2.99/272

For either smaller box, increasing q reduces the iteration count, yet even $q = 448$ is slower than the full box with $q = 192$. On the full grid, increasing q from 192 to 448 reduces the iterations from 794 to 272, but total time is lowest at $q = 384$. Thus box size and rank are not interchangeable, and the rank giving the fewest iterations need not give the lowest time. The two fastest rows come from single runs, so their small difference does not establish a universal best rank.

C.10. Stage 2 fixed-system timing protocol. For every controlled case, the Fourier weights, Toeplitz generator, right-hand side, and regularization are constructed once and fingerprinted with SHA-256. The fingerprint is checked again after all methods finish. The reported method set is CG, Jacobi, and the proposed full-grid, active-inverse, and active-eigenpair family members, as applicable to the grid size. The method order is reshuffled in every repetition and the eigensolver seed is fixed. A timing repetition reruns score-box selection for every active method, verifies that its frozen box hash, rule, and effective rank are unchanged, rebuilds the method-specific preconditioner, starts from zero, solves, synchronizes the GPU at every timing boundary, and then recomputes the true residual in complex128. A row is eligible only if all claimed repetitions satisfy the 10^{-7} tolerance.

The formal target loads the exact $N = 3 \times 10^7$, noise-0.3 Synthetic artifact used in Stage 1. The default active box has $|B| = 10609$ and rank 320. Method configurations are transferred from the prior sweep and frozen before formal timing; no measured repetition tunes a box, rank, or method. CG, Jacobi, the default, active EigenPro, the full-grid spectral family member, and active inverse are all executed.

The common Fourier construction is excluded from the primary solver total because it is shared by all methods. The compared total is active-set selection plus preconditioner construction plus solve; for CG it is the solve alone. The formal reporter recomputes this sum and convergence from repeat-level files. It verifies equal initial, final, and per-repeat system fingerprints, the timing-system SHA-256, the embedded canonical configuration hash, and the component hashes of the weights, Toeplitz generator, and both right-hand-side representations. It also checks the common tolerance, iteration limit, and zero initial vector before emitting Table 4.5. The formal run passes these checks with system identifier 12c0a9811168...; older build-plus-solve artifacts that omit selection time are not used for that table.

C.11. Manitowoc provenance, capacity, and reproduction. NOAA identifies the independent source as the 641-square-mile Manitowoc work unit 300411 and links its official USGS EPT endpoint. It also records that EPT conversion changes the horizontal CRS to EPSG:3857 without changing NAVD88 GEOID18 elevations [23]. The USGS project report gives the native EPSG:6345 horizontal CRS, EPSG:5703 vertical CRS, and 2023 collection dates [33]. The work-unit report gives a mean classification-2 density of 11.37 points m^{-2} [32].

The frozen bounding square is centered at (437100, 4884400) m in EPSG:6345

and has side 52000 m. It contains the entire conforming work-unit extent. The preprocessor first uses EPT node bounds only as a coarse filter, then transforms every candidate point and applies the exact square test. It keeps finite class-2 rows, assigns a stable source identifier from the node rank and point ordinal, and applies SplitMix64 modulo five (residue zero is test). An independent hash orders rows within each EPT depth; shallower additive levels precede deeper levels. The train/test split is target-blind, disjoint, and without replacement. Both coordinates share the fixed 52-km scale, while a target center and scale fitted on the first one million rows of the canonical training order are reused at every size and future depth. The sidecars verify exact nested 100k, 1M, and 10M prefixes and 2.5 million test rows for the largest artifact.

The EPT header reports 24,781,638,330 source points across all classes. The exact scan through LOD 6 yields 40,787,587 training-bucket and 10,197,897 test-bucket class-2 points. The official area and mean-ground-density figures give a planning estimate of 18,876,273,646 ground points, or 15,101,018,917 training-bucket points after the fixed 80/20 split. This estimate motivates a streamed route to a 300M/75M artifact, but it is not an exact deep-level count. No 300M artifact and no 300M solver run are reported; both require a deeper exact scan, and the present resident-array solver cannot load that task on its 4-GB GPU.

Dataset support was not chosen in a blind held-out evaluation. A first pilot used a fixed 8-km square in the separate Buffalo work unit WI_2County_2_B23. On its $N = 10^7$ matched system, CG takes 5374 iterations and 10.972 seconds, while the default takes 1717 iterations and 5.863 seconds, a $1.9109\times$ paired median speedup; the full-grid family member is faster. That pilot prompted the move to full-work-unit support before Manitowoc was screened. We therefore retain the Buffalo result only here as a negative geometry pilot and classify Manitowoc as exploratory despite its independent acquisition.

From the repository root, the checked preprocessing command is:

```
PRE=efgp_eigenpro.py.gpu.benchmark_dataset.preprocess_usgs_ept_wi_2county
EPT=https://s3-us-west-2.amazonaws.com/usgs-lidar-public
PREFIX=USGS_EPT_WI_2County_1_B23
python -m "$PRE" \
  --ept-json "$EPT/WI_2County_1_B23/ept.json" \
  --n-train-list 100000,1000000,10000000 \
  --aoi-center-x 437100 --aoi-center-y 4884400 --aoi-side-m 52000 \
  --aoi-crs EPSG:6345 --max-lod-depth 6 \
  --source-project WI_2County_1_B23 \
  --official-mean-ground-density 11.37 \
  --official-work-unit-area-km2 1660.1823787253759 \
  --dataset-stem-prefix "${PREFIX}_full_workunit_ground_elevation"
```

The matched timing and separate prediction audit are reproduced by:

```
CTL=efgp_eigenpro.py.gpu.box.toeplitz_active_block.controlled
STEM=${PREFIX}_full_workunit_ground_elevation_n10000000
OUT=efgp_eigenpro.py/gpu/box/toeplitz_active_block/controlled/outputs
RUN=${OUT}/integrated_manitowoc_n10000000_matern_q256_primary
python -m "$CTL.benchmark" \
  --dataset-stem "$STEM" \
  --n-train 0 --kernel matern --lengthscale 0.1 --nu 1.5 \
  --variance 1 --lambda 0.1 --fourier-eps 1e-5 --nufft-tol 1e-10 \
  --tol 1e-7 --maxiter 6000 --precision fp64 \
  --methods cg,jacobi,default,full-eig \
  --score-tau 1 --box-budget 8192 --inverse-max-size 1024 \
  --rank 256 \
  --eig-tol 1e-3 --eig-maxiter 1280 --warmup-repeats 1 \
  --measured-repeats 5 --method-order-seed 20260823 \
  --eig-seed 0 \
  --nufft-backend none --precompute-chunk-size 1000000 \
  --post-diagnostic-mode cheap --strict-gpu-eig \
  --output-dir "$RUN"
```

```
python -m "$CTL.prediction_audit" \
--config "$RUN/experiment_config.json" \
--prediction-chunk-size 100000 --warmup-solves 1 \
--output-dir "$RUN/prediction_audit"
```

C.12. Fixed-system bridge to legacy configurations. Table C.8 revisits the archived rank and box sizes without choosing the fastest row after timing. The Matérn columns form the rank-128 configuration bridge and use $|B| = 2601$. The SE columns are a separate matched full-inverse control: their score threshold gives $|B| = 841$, and they also add the archived full grid $|J_m| = 1225$. Full inversion is not formed for the Matérn grid with $M = 35721$. Every displayed method converges in all five measured repetitions.

TABLE C.8

Matched archived-configuration bridge. Each entry is median iterations / old-runner construction-plus-solve seconds / paired speedup over CG; active-method score selection is not included. Entries in different columns are separate fixed-system cases and are not combined.

Method	Synthetic Matérn	USGS Matérn	Synthetic SE	USGS SE
CG	4246/8.657/1.000	4905/10.158/1.000	2054/3.279/1.000	1528/2.458/1.000
Jacobi	1716/4.125/2.111	8447/20.509/0.497	1724/3.262/0.996	12039/23.711/0.103
$P_{\text{inv}}(B)$	169/2.514/3.464	1180/5.667/1.793	6/0.159/20.677	12/0.162/15.069
$P_{\text{inv}}(J_m)$	—	—	1/0.289/11.356	1/0.314/7.853
$P_{\text{eig}}(B, 128)$	733/2.465/3.522	1488/4.622/2.198	156/0.705/4.643	12/0.373/6.630
$P_{\text{eig}}(J_m, 128)$	704/3.144/2.754	303/1.846/5.528	106/0.586/5.600	3/0.359/6.803

For SE, full-grid inversion reduces PCG to one iteration, but its median build cost is 0.285 seconds on Synthetic and 0.310 seconds on USGS. The active inverse builds in 0.144 and 0.135 seconds, respectively, and is therefore $1.82\times$ and $1.94\times$ faster in cold time than the full inverse. This directly demonstrates why the archived fastest-row rule cannot be replaced by iteration count alone.

C.13. Untimed prediction audit. The controlled prediction audit is deliberately separate from the timing repetitions. It rebuilds one system per data set, solves every reported Stage 2 method against that shared system, rechecks the fingerprint, loads the same held-out test artifact, and evaluates the prescribed Fourier predictor in chunks of 100000 points. Coefficients are transferred to the GPU once per method. Neither the audit solve nor prediction time enters a speedup. The audit CSV records the system hash, independently recomputed residual, test RMSE, RMSE ratio and difference relative to CG, prediction time, test size, and chunk size.

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